

Activity: Apply more filters in SQL

Scenario

In this scenario, you're investigating a recent security incident.

You need to gather information about login attempts for certain dates and times. This will help in resolving a security incident.

Task 1. Retrieve login attempts after a certain date

In this task, you need to investigate a recent security incident. To do this, you need to gather information about login attempts made after a certain date.

1. Complete the SQL query to retrieve data for login attempts made after '2022-05-09'. Replace X with the correct operator:

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date > '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1



How many login attempts were made after 2022-05-09?

☐ 134

☐ 185

☒ 125

☐ 111

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195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1
196	acook	2022-05-10	09:56:48	CAN	192.168.52.90	0
198	yappiah	2022-05-12	10:37:22	MEXICO	192.168.103.106	1
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0
200	jclark	2022-05-12	01:11:45	CANADA	192.168.91.103	1

125 rows in set (0.063 sec)

2. Complete the SQL query to retrieve data for login attempts that were made on or after '2022-05-09'. Replace X with the correct operator:

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date >= '2022-05-09';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1

How many login attempts were made on or after 2022-05-09?

☒ 165

☐ 190

☐ 186

☐ 143

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195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1
196	acook	2022-05-10	09:56:48	CAN	192.168.52.90	0
198	yappiah	2022-05-12	10:37:22	MEXICO	192.168.103.106	1
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0
200	jclark	2022-05-12	01:11:45	CANADA	192.168.91.103	1

165 rows in set (0.001 sec)

Task 2. Retrieve logins in a date range

In this task, you need to narrow the focus of the search. Login attempts made after 2022-05-11 shouldn't be included. Use the `BETWEEN` and `AND` operators to return results between '2022-05-09' and '2022-05-11'.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_date BETWEEN '2022-05-09' AND '2022-05-11';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0



How many login attempts were made between 2022-05-09 and 2022-05-11?

☒ 123

☐ 160

☐ 157

☐ 134

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188	jsoto	2022-05-11	00:39:09	USA	192.168.21.88	0
190	jsoto	2022-05-09	05:09:21	USA	192.168.25.60	0
192	bisles	2022-05-10	08:32:03	USA	192.168.201.40	1
195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1
196	acook	2022-05-10	09:56:48	CAN	192.168.52.90	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

123 rows in set (0.001 sec)

Task 3. Investigate logins at certain times

In this task, you need to investigate logins that were made at certain times. To do this, filter the data in the `log_in_attempts` table by login time (`login_time`).

First, your organization's typical work hours begin at 07:00:00. Retrieve all login attempts made before 07:00:00 to learn more about the users who are logging in outside of typical hours.

1. Write a SQL query to retrieve data for login attempts made before '07:00:00'.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time < '07:00:00';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0

What is the username of the fifth record returned from this query?

☐ bisles

☐ acook

☒ eraab

☒ jrafael

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I got it wrong as I didn't notice that the event_id is not same as record other. Lesson learnt in funny way.

The query in the previous step returned more results than required.

2. Modify the query to return logins between '06:00:00' and '07:00:00'

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_time BETWEEN '06:00:00' AND '07:00:00';
```

event_id	username	login_date	login_time	country	ip_address	success
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1

What time was the earliest login attempt between 06:00:00 and 07:00:00?

☐ 06:15:41

☐ 06:03:41

☒ 06:01:31

☐ 06:04:34

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event_id	username	login_date	login_time	country	ip_address	success
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
98	gesparza	2022-05-11	06:30:14	CANADA	192.168.148.80	0
106	tmitchel	2022-05-12	06:15:41	MEXICO	192.168.3.252	1
134	iuduike	2022-05-09	06:46:40	USA	192.168.22.115	1
136	mabadi	2022-05-10	06:56:44	US	192.168.214.234	1
142	gesparza	2022-05-11	06:31:14	CANADA	192.168.117.56	1
147	yappiah	2022-05-08	06:04:34	MEX	192.168.65.245	0
148	daquino	2022-05-08	06:15:55	CANADA	192.168.135.6	1
182	lyamamot	2022-05-10	06:01:31	USA	192.168.106.52	0
191	cjackson	2022-05-08	06:46:07	CANADA	192.168.7.187	0
195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1

15 rows in set (0.001 sec)

Task 4. Investigate logins by event ID

In this task, you need to investigate login attempts based on event ID numbers. With this query, you want to return only the `event_id`, `username`, and `login_date` fields from the `log_in_attempts` table.

Note: The `event_id` column contains numeric data; do not place numeric data in quotation marks.

1. Write a query to return login attempts with `event_id` greater than or equal to 100.

```
MariaDB [organization]> SELECT event_id, username, login_date  
-> FROM log_in_attempts  
-> WHERE event_id >= 100;
```

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11



What is the login date of the third result returned by your query?

☐ 2022-05-08

☒ 2022-05-09

☐ 2022-05-11

☐ 2022-05-10

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The query in the previous step returned more data than required.

2. Modify the query to return only login attempts with `event_id` between 100 and 150.

```
MariaDB [organization]> SELECT event_id, username, login_date  
->  
-> FROM log_in_attempts  
->  
-> WHERE event_id BETWEEN 100 AND 150;
```

event_id	username	login_date
100	tmitchel	2022-05-12
101	sbaelish	2022-05-08
102	jreckley	2022-05-09
103	jhill	2022-05-11
104	asundara	2022-05-11
105	cjackson	2022-05-12
106	tmitchel	2022-05-12
107	bisles	2022-05-12
108	daquino	2022-05-09

What is the username of the seventh result returned by your query?

☐ gesparza

☐ bisles

☐ mabadi

☒ tmitchel

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